CSE400 Fundamentals of Probability in Computing

Domain : Finance

Group 1 Members:

- Kush Kelaiya AU2440018
- Kesar Thakkar AU2420041
- Ishika Patel AU2440088
- Pranav Jain AU2420166
- Nissarg Danidhariya AU2440261

Project Context and Motivation

Project title:

Deterministic Portfolio Optimization for Risk-Aware Investment Allocation

Application Domain:

Financial Portfolio Management

Why Uncertainty Exists:

- Stock returns fluctuate due to market behavior
- Economic events influence asset performance
- Dependencies between assets affect portfolio risk

Problem Context:

Investors must allocate capital among multiple assets while managing uncertainty in future returns.

Project Objective and Deterministic Baseline

Project Objective:

Determine optimal portfolio weights that maximize expected return while controlling risk and diversification.

Deterministic Baseline Strategy:

- Use historical data to estimate return statistics
- Construct a deterministic optimization model
- Compute portfolio weights using fixed optimization rules
- Decision rule is deterministic for given inputs

Why This Baseline is Reasonable:

- Based on classical portfolio theory (Markowitz model)
- Widely used in quantitative finance
- Provides interpretable risk-return tradeoff

System Pipeline Overview

Simulation Pipeline:

1. Historical market data collection
2. Return computation from price series
3. Statistical parameter estimation
4. Deterministic portfolio optimization
5. Risk and performance evaluation

Data Flow:

Market Data → Return Estimation → Statistical Modeling → Deterministic Optimization → Optimal Portfolio and Risk Evaluation

Where Uncertainty Enters:

Uncertainty arises in the asset return data used as input to the model.

Sources of Uncertainty

Market Variability:

- Price fluctuations caused by supply and demand
- Economic and geopolitical influences
- Volatility in financial markets

Data Uncertainty:

- Limited historical data
- Measurement noise in market prices
- Estimation errors in statistical parameters

Dependency Uncertainty:

- Asset returns are correlated
- Correlations may change over time
- Sector-wide events influence multiple assets simultaneously

Key Random Variables

- **Asset Return of asset i :** $R_i = (P[t+1] - P[t])/P[t]$

- **Portfolio Return:** $R_p = w` R$
  Weighted combination of asset returns

- **Portfolio Loss:** $L_p = -R_p$
  Represents negative outcomes in the portfolio

- **Distributional Assumption:** $R \sim N(\mu, \Sigma)$

Deterministic Baseline Logic

- **Return Model:** $R \sim N(\mu, \Sigma)$
- **Portfolio Return:** $R_p = w' R$
- **Expected Portfolio Return:** $\mu_p = w' \mu$
- **Portfolio Risk:** $p^2 = w'\Sigma w$
- **Deterministic objective:** $\max_w (\mu_p - \lambda\sigma_p)$

Deterministic Baseline Logic

- The Idea: We use a set-in-stone rule to decide which stocks to buy.
- The Goal: To have a standard starting point to compare with more advanced methods later.
- How it works(Steps) :
  1. Checking the Data: We look at the expected profit and risk for each stock.
  2. Find the probability : We calculate the percentage that a stock will make money (profit).
  3. The Decision: If the probability of profit is 50% or higher, the system picks that stock.
- No Randomness: The system follows this exact same rule every single time and no random things.

Simulation Implementation

Algorithms Used:

- Sharpe Ratio Optimization: A method used to find the best balance between high returns and low risk.
- SLSQP Solver: A mathematical tool which calculates the best investment based on weights without guessing.
- Markowitz Portfolio Theory: It is used to visualize the "Efficient Frontier"(the best portfolio).

Code Structure Overview:

- Data Retrieval: The script uses yfinance to download real-world historical prices for stocks like AAPL and MSFT.
- Statistical Math: It calculates the average annual return and the risk for each asset.
- Optimization Functions: Code defines specific functions to calculate the portfolio's total return, risk, and Sharpe Ratio.

Full Pipeline Execution:

1. Input: Historical data is fed into the model.
2. Processing: The model analyzes how these stocks move together and finds the Optimal Weights.
3. Simulation: It creates 5,000 random portfolios to show all the possible choices.
4. Output: It picks the single best Deterministic Optimal Portfolio and marks it on a graph.

Result and Performance Analysis

The "Markowitz Bullet" Visual

- Random Portfolios : We tested 5,000 different ways to mix our stocks. Each dot represents a different possible portfolio.
- The Efficient Frontier: The curved shape of the dots shows that as we want more profit then we have to take on more risk.
- The "X" Mark: Our deterministic code successfully found the single best point on this curve—the portfolio with the highest return for the lowest risk.

Key Performance Metrics

Based on our simulation using AAPL, MSFT, GOOGL, and AMZN, we achieved the following baseline results:

- Expected Annual Return: This is the average profit our strategy expects to make in a year (e.g., 25.5%).
- Volatility (Risk): This shows how much the price might swing up or down (e.g., 18.2%).
- Sharpe Ratio: A score that tells us if our profit is worth the risk. A higher score means a better, "smarter" investment.

Observed Behavior

- **Consistency**: Because we used a fixed math rule (SLSQP), the model does not guess. It always picks the mathematically "safest" way to reach high returns.
- **Stability**: The deterministic baseline avoids "crazy" or "gambling" investments, staying strictly on the edge of the Efficient Frontier.

Limitations of Deterministic Strategy

- **Fixed and Rigid**: Because the rule is "deterministic" (fixed), it cannot change. If the market suddenly crashes or changes, the model keeps following the same old math, which can lead to losses.
- **The "Bell Curve" Problem**: Our code assumes stock returns follow a smooth "Normal Distribution" (the Bell Curve). However, real-world markets often have "Black Swan" events—extreme jumps or drops that the math doesn't expect.
- **No Exploration**: The model only looks at what it already knows from the historical data it has . It never "tries out" new or emerging stocks that might earn high profit in the future.
- **Ignoring Connections**: The strategy looks at stocks mostly on their own. In reality, when one big stock drops, others small ones also drop with it. A fixed rule struggles to handle these connected movements.

Summary and Transition to Randomized Strategies

Key Insights from our Current Model:

- Our "Fixed" (Deterministic) strategy is a strong starting point because it is based on solid math.
- It successfully finds the "best" portfolio if the market stays exactly the same as in the past.
- It gives us a reliable "baseline" score to compare with future models.

Why we need a Randomized Approach next:

- **The Market Changes**: Real-world stock prices are not always predictable. A fixed rule can be too slow to react to sudden changes which happen.
- **The "Exploration" Problem**: Our current model only uses what it already knows. It never "tries out" new stocks to see if they might perform better.
- **Next Step (Milestone 4)**: We will move to Randomized Strategies. This will allow our system to "learn" from its mistakes and adapt to get even better results.